KINGDOM OF SAUDI ARABIA
IMAM MOHAMMAD IBN SAUD ISLAMIC UNIVERSITY
COLLEGE OF COMPUTER AND INFORMATION SCIENCES
INFORMATION SYSTEMS DEPARTMENT
BACHELOR IN INFORMATION SYSTEMS

المملكة العربية السعودية
جامعة الإمام محمد بن سعود الإسلامية
كلية علوم الحاسب والمعلومات
قسم نظم المعلومات
بكالوريوس نظم المعلومات

SYLLABUS

IS1380: Cyber Security

CREDIT HOURS	3 Lecture hours

PREREQUISITE	IS1221

Instructor:
Contact information and office hours
Office No: To be announced (TBA) **Office Hours: TBA** **E-mail: _____@imamu.edu.sa**

COURSE DESCRIPTION
The target of the course is to introduce topics related to cybersecurity. The students will learn about cybersecurity terminologies, basic cryptography techniques, access control authorization and authentication, some direct emerging topics of cybersecurity, malicious software, ethical hacking overview, cybersecurity ethics and laws and cybersecurity systems development project.

COURSE LEARNING OUTCOMES (CLOs)		Aligned SOs
1	**Knowledge and Understanding**	
1.1	Define the fundamental concepts, methods, and practice of cybersecurity.	1(I)
1.2	Outline the key concepts of cybersecurity operating system and strategic defense.	1(I)
1.3	Describe the fundamental issues in designing access control models and authentication.	1(I)

1.4	List the main characteristics of social engineering and malicious software	1(I)
1.5	State cybersecurity regulation, ethics and laws.	4(P)
2	**Skills :**	
2.1	Explain the basic cryptography concepts.	1(I)
2.2	Plan and organize an information systems cybersecurity development project at introductory level.	2(P)
2.3		
2.4		
2.5		
3	**Values:**	
3.1	Function effectively on teams to accomplish a common goal.	5(P)
3.2	Present a topic in a compelling manner.	3(P)
3.3		
3.4		
3.5		

TEACHING Strategies
Lectures Self-Learning

No	List of Topics	Contact Hours	Self-Learning
1	Introduction to cybersecurity	*3*	
2	Basic Cryptography	*3*	
3	Operating System Security	*3*	*2*
4	Strategic Defenses	*3*	*2*
5	Access Control-Authentication	*6*	
6	Access Control-Authorization	*3*	
7	Cybersecurity Emerging topics	*6*	
8	Social Engineering and malware	*3*	*2*
9	Cybersecurity Frameworks, Ethics and Laws	*3*	
10	Project Discussion	*3*	
11			
12			
Total		**36**	**6**

TEXT BOOK

Security in Computing, Fifth Edition; Charles P. Pfleeger Shari Lawrence Pfleeger Jonathan Margulies, Published January, 2015 by Pearson Education;ISBN-10: 0-13-408504-3, ISBN-13: 978-0-13-408504-3

REFERENCES

Understanding Cryptography, 1 edition, ChristofPaar; Jan Pelzl, Springer, 2010, ISBN 978-3642041006.

Computer Security, 3rd edition, Dieter Gollmann, John Wiley&Sons, 2011, ISBN 978-0470741153.

The Basic of Information Security, Jason Andress, 2014 Elsevier Inc, Second Edition

	Course Assessment Methods		
No	**Assessment Method**	**Due Week**	**% of Total Assessment**
1	**Quiz**	4	10
2	**Assignment**	9	10
3	**Midterm**	7	20
4	**Project / Lab Exam**	11	20
5	**Final Exam**	13	40